Class room activity

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Question:

Activity 1: Design a Star Schema (15 Marks)

Question 1: A college wants to develop a data warehouse to analyze student academic performance across different departments, courses, and semesters. Design a Star Schema for this application. Identify the fact table and dimension tables, specify the attributes of each table, and indicate the primary and foreign keys. Draw a neat star schema diagram to represent the relationships among the tables.

Activity 2: Perform ETL Operations (10 Marks)

Question 2: A college collects student information from admission, examination, and attendance systems. The collected data contains missing values, duplicate records, inconsistent department names, and different date formats. Explain and demonstrate the Extract, Transform, and Load (ETL) process required to integrate this data into a data warehouse. Show sample input records, the transformations performed, and the final cleaned data loaded into the target table.

Activity 3: Demonstrate OLAP Operations (15 Marks)

Question 3: Consider a data warehouse containing sales information with the dimensions Time, Product, and Location, along with sales amount as a measure. Construct a sample dataset and demonstrate the following OLAP operations with suitable examples and output tables:
a) Roll-up – Summarize monthly sales into yearly sales.
b) Drill-down – Display yearly sales as month-wise sales.
c) Slice – Retrieve sales data for one specific product.
d) Dice – Retrieve sales data for selected products, locations, and time periods.
e) Pivot – Change the orientation of rows and columns to compare sales across products and locations.

Activity 4: Draw Data Warehouse Architecture (10 Marks)
Question 4: A hospital wants to integrate patient registration, laboratory, pharmacy, and billing data to support management decision-making. Design a Data Warehouse Architecture for the hospital. Draw a neat, labelled diagram showing the data sources, ETL process, staging area, central data warehouse, data marts, OLAP tools, and reporting tools. Briefly explain the function of each component and describe how the architecture supports decision-making.

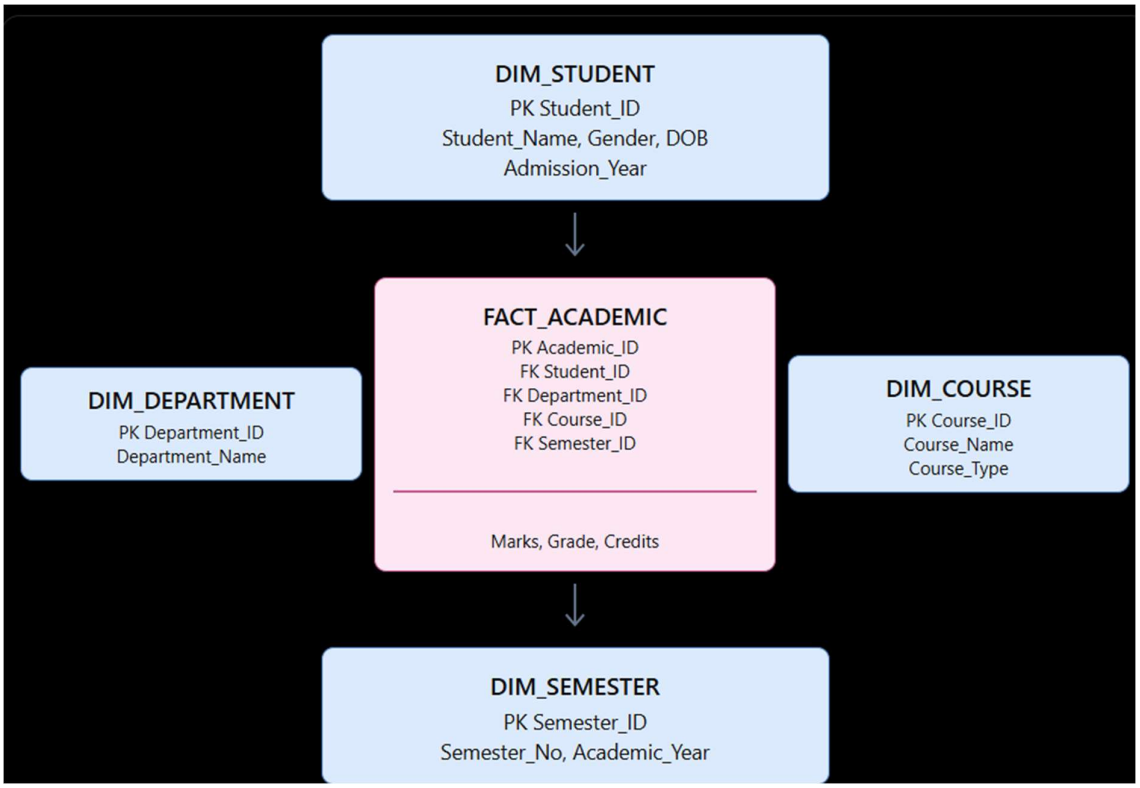
Answers

1-Design a Star Schema:

1. Aim

To design a Star Schema for a college data warehouse to analyze student academic performance across departments, courses, and semesters.

2. Star Schema diagram



3. Fact table

FACT_ACADEMIC stores measurable academic performance information.

Attribute	Key / Description
Academic_ID	PK – unique record ID
Student_ID	FK – references DIM_STUDENT

Department_ID	FK – references DIM_DEPARTMENT
Course_ID	FK – references DIM_COURSE
Semester_ID	FK – references DIM_SEMESTER
Marks	Marks obtained
Grade	Grade achieved
Credits	Course credits earned

4. Dimension tables

DIM_STUDENT

Attribute	Key / Description
Student_ID	PK
Student_Name	Student's name
Gender	Student's gender
DOB	Date of birth
Admission_Year	Year of admission

DIM_DEPARTMENT

Attribute	Key / Description
Department_ID	PK
Department_Name	Department name
Faculty_Name	Faculty or department head

DIM_COURSE

Attribute	Key / Description
Course_ID	PK
Course_Name	Name of the course
Course_Type	Core or elective
Credits	Course credit value

DIM_SEMESTER

Attribute	Key / Description
Semester_ID	PK
Semester_No	Semester number
Academic_Year	Academic year
Start_Date	Semester start date
End_Date	Semester end date

5. Explanation and uses

- The fact table stores academic performance measures such as marks and credits.
- Dimension tables provide descriptive information about students, departments, courses, and semesters.
- Primary keys uniquely identify records in each table.
- Foreign keys establish relationships between the fact table and dimension tables.
- The college can compare average marks by department, identify top-performing courses, and track student performance across semesters.

Conclusion: The Star Schema provides a simple and efficient structure for analyzing academic performance and generating college management reports.

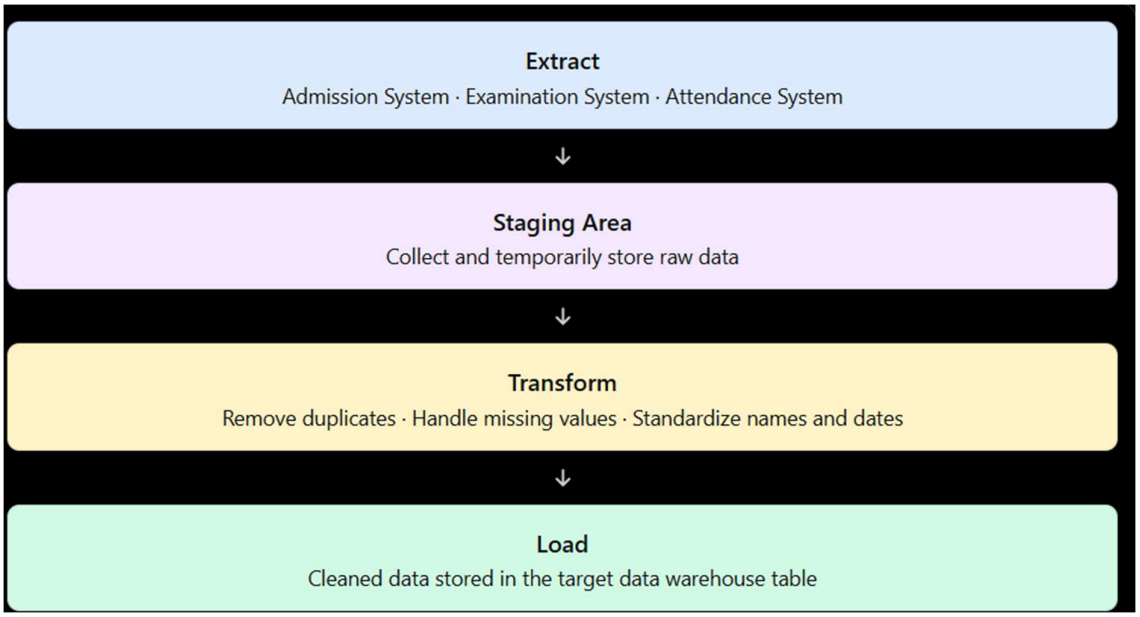
2- Perform ETL Operations:

1. Definition

ETL stands for Extract, Transform, and Load. It is a process used to collect data from different sources, clean and standardize it, and store the prepared data in a data warehouse.

2. ETL process diagram

To design a Star Schema for a college data warehouse to analyze student academic performance across departments, courses, and semesters.



3. Sample input records

Suppose the college collects student information from admission, examination, and attendance systems.

Student_ID	Name	Department	Marks	Attendance	Date
101	Arun	CSE	85	90	#######
102	Banu	Computer Science	NULL	80	#######
101	Arun	CSE	85	90	#######
103	Charan	ECE	72	NULL	#######

Problems identified:

- Missing marks for student 102.
- Missing attendance for student 103.
- Duplicate record for student 101.
- Inconsistent department names such as CSE and Computer Science.
- Different date formats.
- design a Star Schema for a college data warehouse

4. ETL operations

Step 1: Extract

- Collect student details from the admission system.
- Collect marks from the examination system.
- Collect attendance records from the attendance system.
- Combine the extracted records in a staging area.

Step 2: Transform

- Remove duplicate records using Student_ID and the relevant record details.
- Replace Computer Science with the standard department name CSE.
- Convert all dates into the YYYY-MM-DD format.
- Handle missing marks using the examination system's verified result, if available; otherwise, keep the value NULL until it is verified.
- Handle missing attendance using the attendance system's verified data; do not assume an attendance value without evidence.
- Validate records and ensure data types are consistent.

5. Final cleaned data loaded into the target table

Assume the missing values are verified from the original systems before loading.

Student_ID	Name	Department	Marks	Attendance	Date
101	Arun	CSE	85	90	#######
102	Banu	CSE	78	80	#######
103	Charan	ECE	72	85	#######

For demonstration, the missing marks and attendance above are assumed to have been verified as 78 and 85, respectively.

6. Benefits of ETL

- Improves data quality and consistency.
- Removes duplicate records.

- Integrates information from multiple systems.
- Makes data suitable for analysis and reporting.
- Supports reliable decision-making.

Conclusion: ETL integrates data from admission, examination, and attendance systems into a consistent, accurate, and usable data warehouse.

3-Demonstrate OLAP Operations :

1. Definition

OLAP (Online Analytical Processing) is a technology used to analyze multidimensional data and summarize information for decision-making. Here, the dimensions are Time, Product, and Location, and the measure is Sales Amount.

2. Sample dataset

Month	Product	Location	Sales Amount (₹)
Jan-25	Laptop	Chennai	50,000
Feb-25	Laptop	Chennai	60,000
Jan-25	Mobile	Chennai	30,000
Feb-25	Mobile	Chennai	40,000
Jan-25	Laptop	Bangalore	45,000
Feb-25	Laptop	Bangalore	55,000
Jan-25	Mobile	Bangalore	25,000
Feb-25	Mobile	Bangalore	35,000
Jan-26	Laptop	Chennai	70,000
Feb-26	Laptop	Chennai	80,000

(a) Roll-up – Monthly sales into yearly sales

Definition: Roll-up summarizes detailed data into a higher level of aggregation.

Example: Monthly sales are combined to calculate yearly sales for each product and location.

Year	Product	Location	Yearly Sales (₹)
2025	Laptop	Chennai	1,10,000
2025	Mobile	Chennai	70,000
2025	Laptop	Bangalore	1,00,000
2025	Mobile	Bangalore	60,000
2026	Laptop	Chennai	1,50,000
2026	Mobile	Chennai	95,000

Result: Roll-up provides summarized yearly sales instead of month-wise details.

(b) Drill-down – Yearly sales into monthly sales

Definition: Drill-down displays more detailed information by moving from a higher level to a lower level.

Example: The 2025 yearly sales of laptops in Chennai are ₹1,10,000.

Year	Month	Product	Location	Sales (₹)
2025	January	Laptop	Chennai	50,000
2025	February	Laptop	Chennai	60,000
Total				1,10,000

Result: Drill-down helps identify monthly changes in sales.

(c) Slice – Sales data for one product

Definition: Slice selects one specific value from a dimension to produce a smaller data view.

Example: Select only the Laptop product.

Month	Product	Location	Sales (₹)
Jan-25	Laptop	Chennai	50,000
Feb-25	Laptop	Chennai	60,000
Jan-25	Laptop	Bangalore	45,000
Feb-25	Laptop	Bangalore	55,000
Jan-26	Laptop	Chennai	70,000
Feb-26	Laptop	Chennai	80,000

Result: Slice displays sales information for laptops only.

(d) Dice – Sales data for selected products, locations, and time periods

Definition: Dice selects specific values or ranges from two or more dimensions to create a smaller multidimensional dataset.

Example: Select Laptop and Mobile, Chennai, and January–February 2025.

Month	Product	Location	Sales (₹)
Jan-25	Laptop	Chennai	50,000
Feb-25	Laptop	Chennai	60,000
Jan-25	Mobile	Chennai	30,000
Feb-25	Mobile	Chennai	40,000

Result: Dice retrieves sales records that satisfy the selected product, location, and time conditions.

(e) Pivot – Change rows and columns

Definition: Pivot changes the orientation of a data table to provide a different view of the same information.

Example: Compare product sales across Chennai and Bangalore during January 2025.

Product	Chennai Sales (₹)	Bangalore Sales (₹)
Laptop	50,000	45,000
Mobile	30,000	25,000

Result: Pivot makes it easier to compare the sales of different products across locations.

Conclusion

OLAP operations help analyze large volumes of data from different perspectives. Roll-up summarizes data, drill-down provides details, slice selects one dimension value, dice filters multiple dimensions, and pivot changes the presentation of data for comparison.

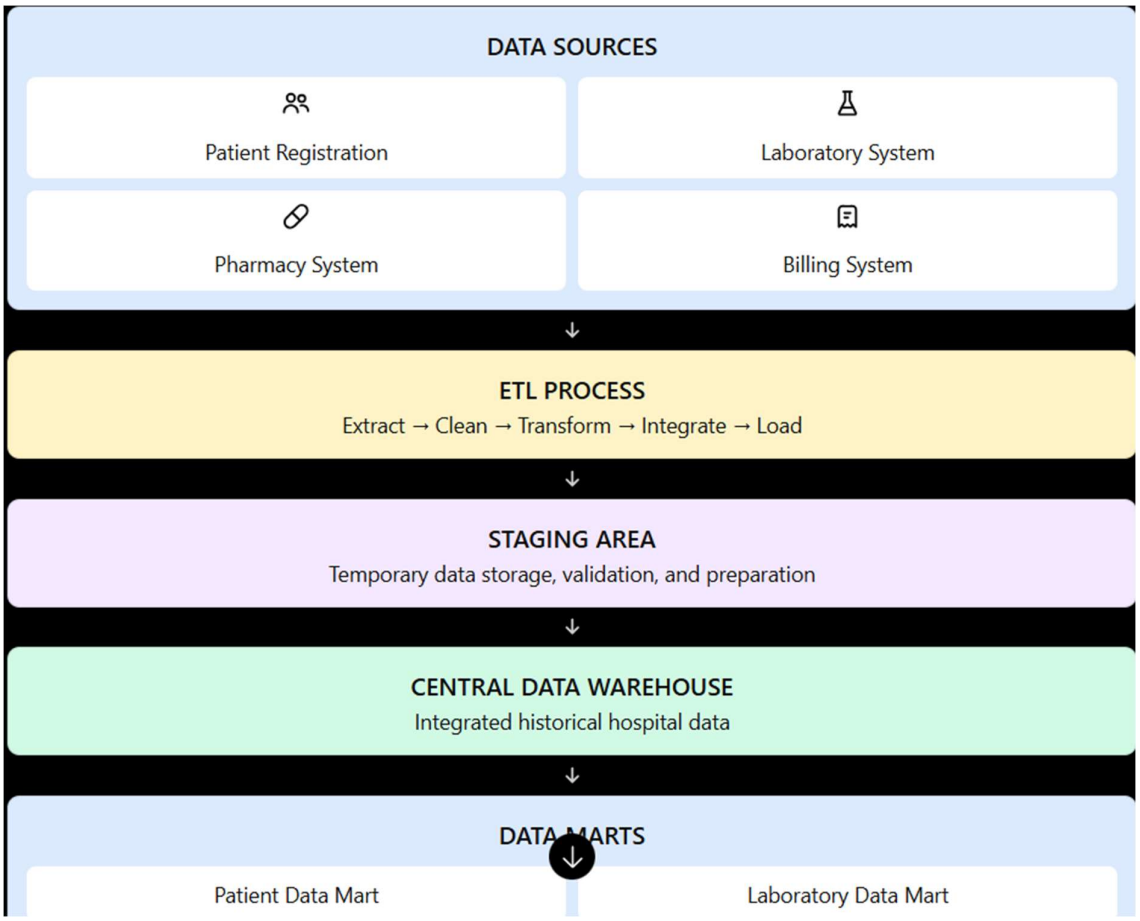
4-Draw Data Warehouse Architecture :

1. Aim

To design a hospital data warehouse that integrates patient registration, laboratory, pharmacy, and billing data to support management decision-making.

2. Data warehouse architecture diagram

OLAP operations help analyze large volumes of data from different perspectives. Roll-up summarizes data, drill-down provides details, slice selects one dimension value, dice filters multiple dimensions, and pivot changes the presentation of data for comparison. Handle missing attendance using the attendance system's verified data; do not assume an attendance value without evidence. hospital data warehouse that integrates patient registration, laboratory, pharmacy, and billing data to support management decision-making.



3. Functions of each component

1. Data sources

Patient registration, laboratory, pharmacy, and billing systems provide the original hospital data. Each system maintains information relevant to its activities.

2. ETL process

ETL extracts data from different systems, cleans missing or inconsistent values, standardizes formats, and integrates the data before loading it into the warehouse.

3. Staging area

The staging area temporarily stores extracted data. It supports data cleaning, validation, and preparation before the data is loaded into the central warehouse.

4. Central data warehouse

The central data warehouse stores integrated, consistent, historical hospital data. It allows management to analyze information across multiple departments.

5. Data marts

Data marts organize data for specific departments, such as patient services, laboratory operations, pharmacy inventory, and billing. They make department-specific analysis easier.

6. OLAP tools

OLAP tools support multidimensional analysis, such as comparing monthly billing totals, analyzing laboratory test trends, and studying pharmacy usage.

7. Reporting tools

Reporting tools generate dashboards, charts, summaries, and reports that help hospital managers monitor performance and make decisions.

4. How the architecture supports decision-making

- Helps management analyze patient numbers and service usage.
- Identifies trends in laboratory tests and pharmacy demand.
- Supports billing analysis and revenue monitoring.
- Reduces inconsistent information across departments.
- Provides historical data for planning staff, medicines, and hospital resources.

Conclusion

The hospital data warehouse integrates data from multiple systems into a central repository. Data marts, OLAP tools, and reporting tools convert this integrated data into useful information for effective planning and management decision-making.